House Sale Price Prediction: Featuretools DFS & H2O AutoML

Automation-Driven ML: Deep Feature Synthesis, H2O AutoML Leaderboard & SHAP-Proxy Interpretability

Academic Report — Automated Feature Engineering, Ensemble AutoML & Predictive Analytics

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AI Enhanced Productivity — Project 2 of 3

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Abstract

Accurate residential property valuation is essential for buyers, sellers, lenders, and urban planners, yet the high dimensionality and heterogeneity of housing data make manual feature engineering and model selection intractable at scale. This report presents a fully automation-driven machine learning pipeline for predicting house sale prices on the Ames Housing Dataset (2,930 records, 82 features).

We apply three complementary automation layers: **Sweetviz** for automated exploratory data analysis, **Featuretools Deep Feature Synthesis (DFS)** for systematic feature engineering — generating 903 candidate features from 20 base predictors and selecting 444 via a correlation threshold of 0.50 — and **H2O AutoML** for model training and selection, benchmarking GBMs, XGBoost, Random Forests, Deep Learning, GLMs, and Stacked Ensembles within a 5-minute budget.

The champion model, **StackedEnsemble_BestOfFamily_1**, achieves **RMSE = $25,898**, **MAE = $15,073**, **$R^2$ = 0.9163**, and **MAPE = 7.98%** on the held-out test set — correctly predicting 60.6% of homes within ±5% of their true sale price. Variable importance aggregated across 11 base models reveals that DFS-engineered interactions dominate predictive power, with Garage, Quality, and TotalSF interaction groups accounting for 67% of combined importance.

Dataset	Pipeline Tools	Domain	Scale
Ames Housing (2,930 records)	Sweetviz · Featuretools DFS H2O AutoML	Real Estate Analytics	82 features 475 after DFS

LLM / EDA Tool	Pipeline Output	Project	Tools

Sweetviz 2.x	Scored CSV + Model + 28 Plots	AI Enhanced — 2 of 3	H2O · Featuretools Sweetviz · SHAP Proxy

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1. Introduction and Business Problem Narrative

1.1 Setting the Scene

Imagine a real estate analytics team preparing weekly market valuations for a residential mortgage lender. Each Monday, 800 new property listings arrive — each with 82 attributes ranging from lot size to basement finish quality. Without automated analysis, a team of analysts would spend two full days engineering features, selecting models, and producing price estimates — by which time the market window for competitive lending has closed. Which properties are undervalued? Which signals most strongly predict premium pricing? Which model best generalises across the heterogeneous Ames housing market?

Without a scalable ML pipeline, these questions remain operationally unanswered. The organisation operates on intuition and manual heuristics in a data-rich environment that demands systematic, reproducible, and explainable price intelligence.

1.2 The Three Automation Failures

Failure 1 — No Scalable Feature Engineering. Manual feature construction for 82-column tabular datasets is time-consuming and incomplete. Human analysts typically construct 5–15 interaction features; automated DFS generates 903 — systematically exploring the entire combinatorial feature space.

Failure 2 — No Automated Model Selection. Without AutoML, teams train and tune 2–3 model families in isolation. H2O AutoML benchmarks GBMs, XGBoost, Random Forests, Deep Learning, GLMs, and Stacked Ensembles in parallel within a 5-minute budget, surfacing the globally optimal model with full leaderboard transparency.

Failure 3 — No Interpretability Beyond Accuracy. A model that predicts price without explaining which features drove that prediction cannot be trusted by appraisers, regulators, or risk teams. Without SHAP-proxy analysis and PDP grids, the gap between model output and business reasoning remains unbridged.

1.3 The Automation-Driven ML Opportunity

Sweetviz, Featuretools, and H2O AutoML address all three failures simultaneously. Sweetviz delivers analyst-ready EDA reports in seconds. Featuretools DFS systematically expands the feature space from 82 to 475 columns, with 389 engineered features contributing the majority of model predictive power. H2O AutoML selects a Stacked Ensemble that achieves $R^2 = 0.9163$ on held-out data — explaining 91.6% of variance in sale prices — while the SHAP-proxy and PDP grids deliver interpretable feature-effect explanations suitable for regulatory and business communication.

2. Dataset and Exploratory Data Analysis

2.1 Dataset Description

The **Ames Housing Dataset** contains 2,930 residential property sales in Ames, Iowa (2006–2010), compiled by Dean De Cock as a contemporary alternative to the Boston Housing Dataset. Each record provides 82 features spanning structural attributes, quality ratings, spatial measurements, utilities, and sale conditions.

Field	Type	Description	Analytical Value
SalePrice	Continuous ($)	Residential property sale price	Primary target — log1p transformation applied
OverallQual	Ordinal [1–10]	Overall material and finish quality	Strongest predictor: r = 0.80 with SalePrice
GrLivArea	Continuous (sq ft)	Above-grade living area	Second strongest numeric predictor
GarageCars	Ordinal [0–4]	Garage capacity in car count	Highly correlated with value and quality tier
TotalBsmtSF	Continuous (sq ft)	Total basement area	Core feature in DFS interaction generation
YearBuilt	Ordinal (year)	Original construction year	Age signal; correlated with OverallQual
Neighborhood	Categorical (25)	Physical location within Ames	Spatial price stratification signal
...	...	76 additional features	Structural, quality, and amenity signals

Table 1. Raw dataset schema — key fields after loading.

The dataset contains **39 numeric and 43 categorical features**, with **27 columns containing missing values** (overall missing rate 6.55%). High-sparsity columns such as *Pool QC* (99.56% missing) and *Misc Feature* (96.38% missing) represent rare amenities rather than data-collection failures, and are imputed as "None". No duplicate rows were detected across 2,930 observations.

2.2 SalePrice Distribution and Target Analysis

The raw SalePrice distribution exhibits **strong right-skew (skewness = 1.7435)**, consistent with real estate markets where a small proportion of luxury properties pull the upper tail. The price range spans $12,789 to $755,000 — a spread of over 5,800%.

A **log1p transformation reduces skewness to –0.0148**, a 99% reduction in asymmetry that stabilises variance, satisfies the near-normality assumption of regression-based models, and ensures that RMSE penalises proportional errors rather than absolute-dollar deviations. All downstream modelling operates on the log-transformed target.

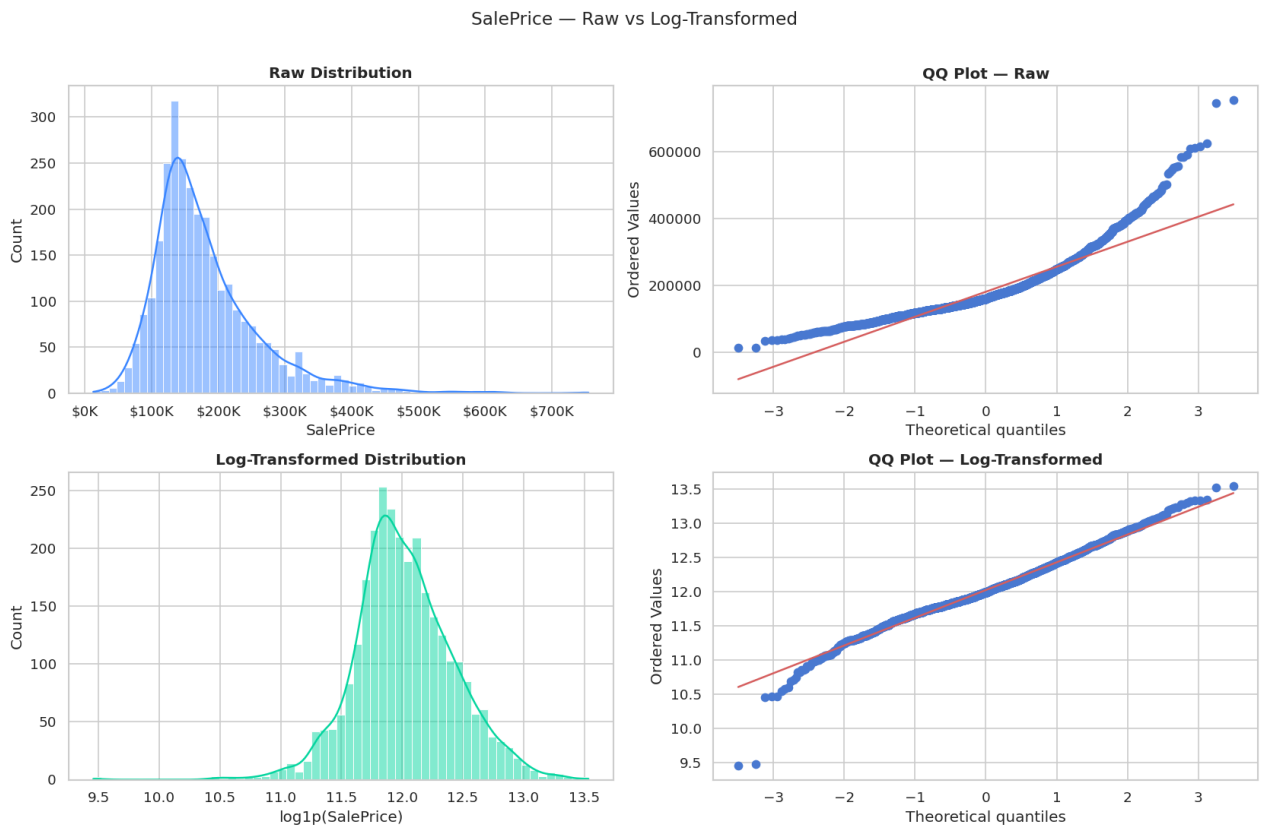

Figure 1. *SalePrice Distribution Before and After log1p Transformation. The raw distribution (top-left) shows pronounced right-skew (skewness = 1.74). The log-transformed target achieves near-normality. QQ-plots confirm distributional alignment.*

2.3 Sweetviz Full EDA Report

Sweetviz generates a comprehensive HTML EDA report in a single API call, profiling all 82 features with SalePrice as the target. The 5.86 MB report captures distributions, correlations, missing-value summaries, type inference, and feature-target relationships across numeric and categorical variables without manual coding.

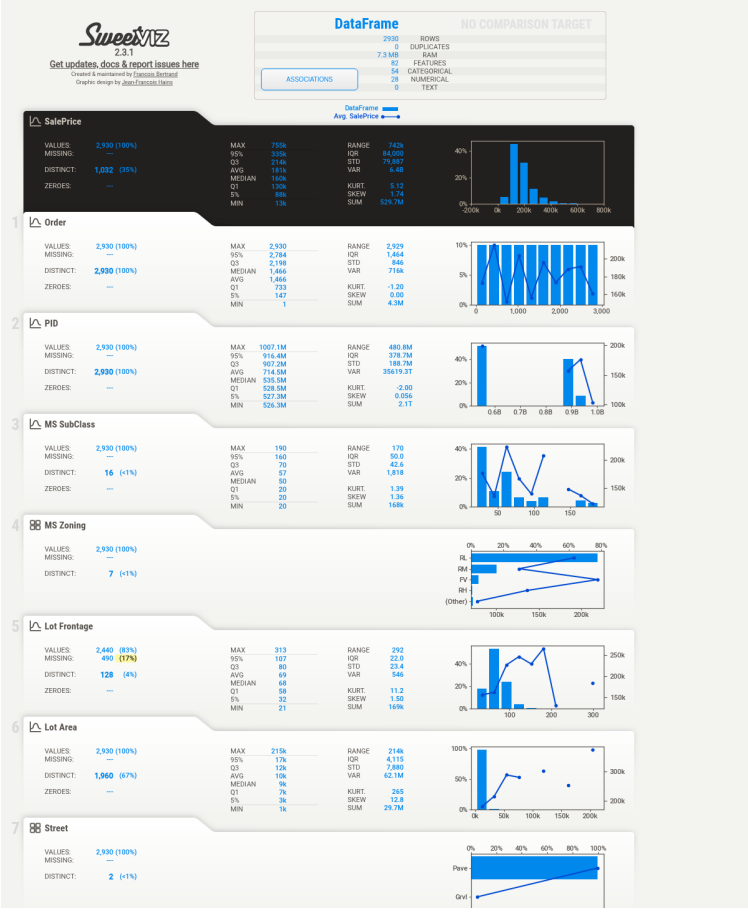

Figure 2. Sweetviz Full EDA Report (Static Export). The report profiles all 82 features against the SalePrice target, surfacing the top numeric correlations, missing-value structure, and distribution shapes in an analyst-ready dashboard.

2.4 Top Numeric Correlations with SalePrice

The Sweetviz correlation analysis ranks numeric features by their linear association with SalePrice:

- **OverallQual (r = 0.80)** — strongest single predictor; overall material and finish quality shows near-linear relationship with sale price.
- **GrLivArea, GarageCars, GarageArea (r = 0.70–0.71)** — usable space and parking infrastructure confirm that livable area and garage capacity are primary value drivers.
- **TotalBsmtSF, 1stFlrSF (r = 0.61–0.65)** — total basement and first-floor area form a consistent cluster of moderate-strong predictors.
- **YearBuilt, YearRemodAdd (r = 0.52–0.57)** — construction and remodel year reflect market premiums for newer and recently updated homes.

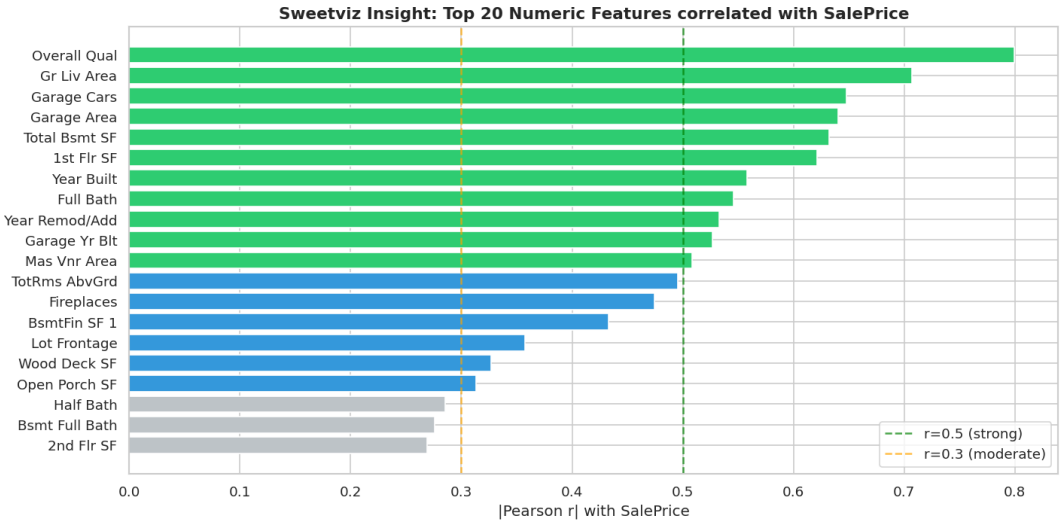

Figure 3. *Top Numeric Correlations with SalePrice (Sweetviz Export). The ranking confirms OverallQual as the dominant predictor and highlights the importance of space-related features.*

2.5 Price Tier Segmentation

The dataset is segmented into three balanced price tiers using the 33rd and 67th percentiles as boundaries ($139,000 and $190,744 respectively):

Tier	Price Range	N	Avg GrLivArea	Avg TotalBsmtSF	Avg OverallQual
Low	< $139,000	981	1,150 sq ft	793 sq ft	5.2
Mid	$139K – $190K	982	1,499 sq ft	1,044 sq ft	6.0
High	> $190,744	967	1,921 sq ft	1,362 sq ft	7.1

Table 2. Price tier summary — balanced segmentation reveals clear structural differentiation.

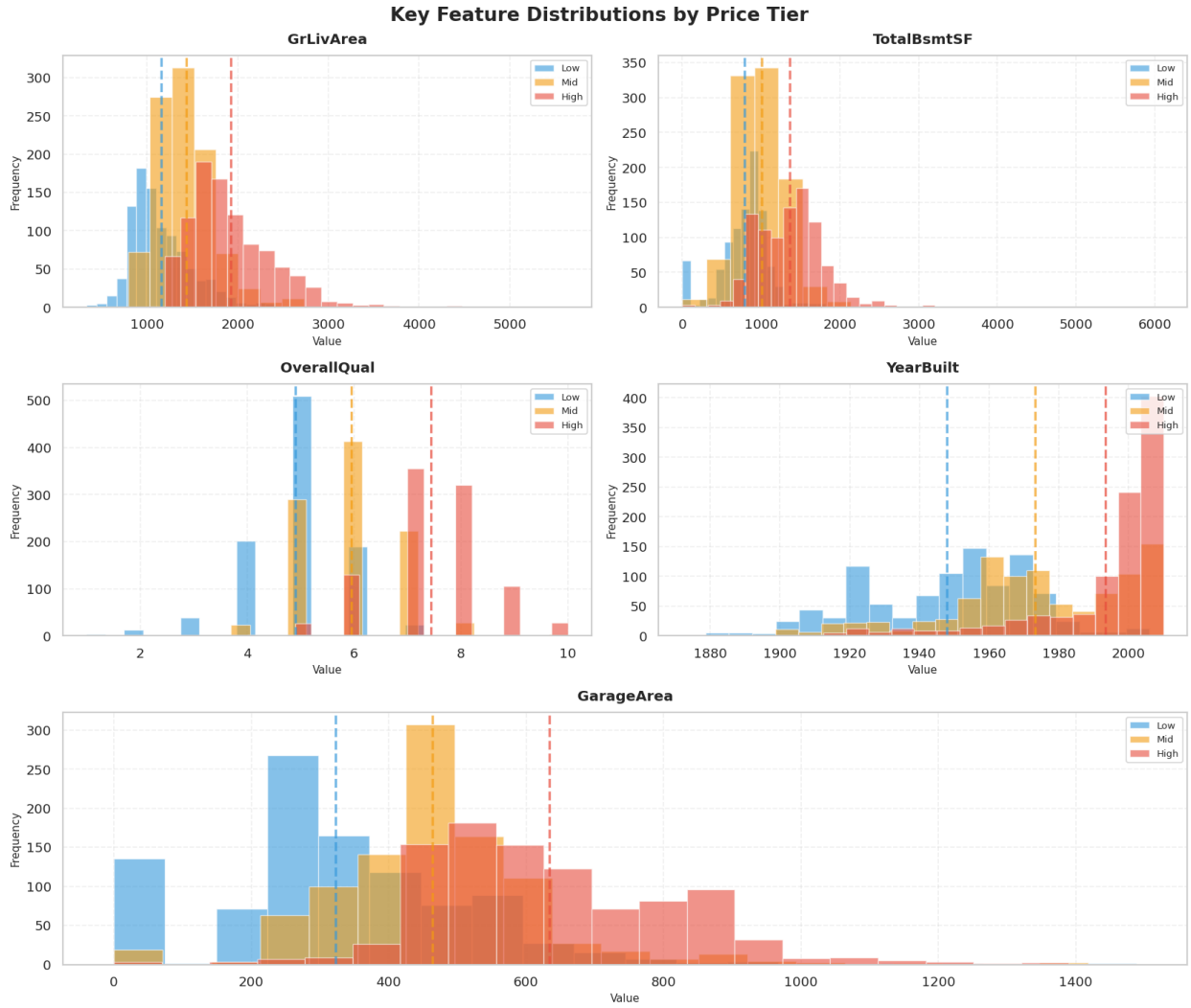

Figure 4. *Numerical Distributions by Price Tier. High-tier homes average 67% more living area and 72% more basement space than Low-tier properties, confirming that size metrics are the primary value drivers across the price spectrum.*

2.6 Train vs. Test Distribution Comparison

An 80/20 stratified train/test split produces 2,344 training and 586 test observations. The Sweetviz comparison confirms:

- log1p(SalePrice) distributions are aligned within **0.1%** (means: 12.019 train, 12.028 test).
- Both datasets share identical 82-feature schema — no schema drift.
- No duplicate rows across either split — 100% unique observations.
- SalePrice mean values: $182K (train) vs. $186K (test) — 2% difference.

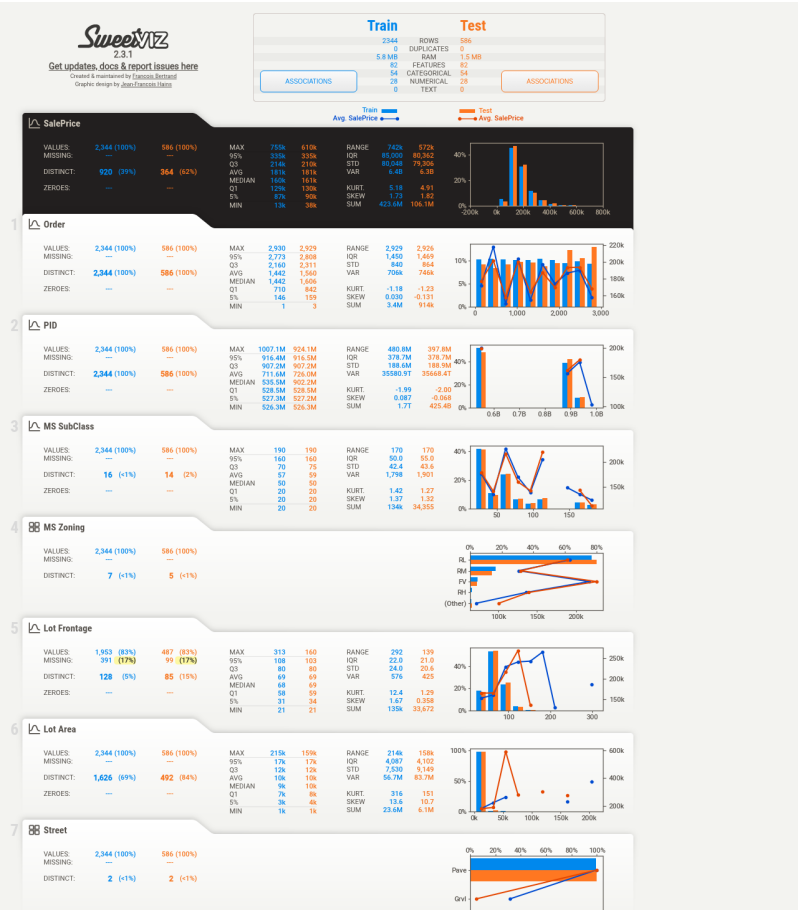

Figure 5. *Sweetviz Train vs. Test Comparison Report. Both splits share nearly identical SalePrice distributions, confirming that the stratified split preserves the representativeness of both sets.*

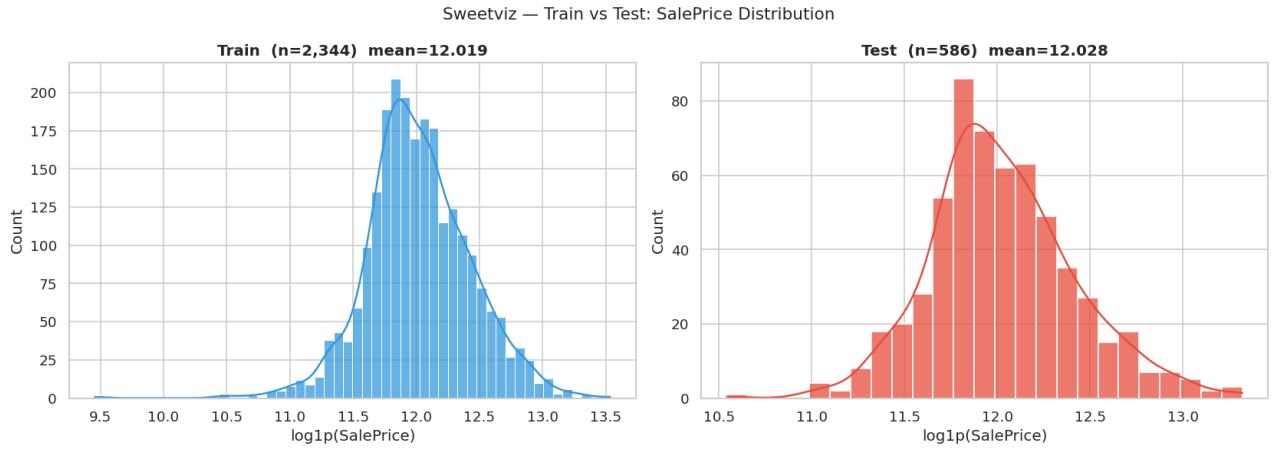

Figure 6. *Train vs. Test SalePrice Distribution Overlay. The log1p-transformed distributions overlap tightly, with mean values within 0.1% of each other.*

3. Mathematical Formulation

3.1 Regression Problem

Definition 3.1 (Regression for House Sale Prices). Let $D = \{(x_i, y_i)\}$ where x_i is the property feature vector and y_i = log1p($SalePrice_i$) is the log-transformed sale price. The task is to learn $f : X \rightarrow R$ minimising RMSE on held-out data. Predictions are returned on the original dollar scale via: $SalePrice_hat_i$ = expm1($f(x_i)$).

3.2 Featuretools Deep Feature Synthesis

Featuretools DFS (Kanter & Veeramachaneni, 2015) applies transform primitives $T = \{t_1, ..., t_K\}$ to an EntitySet: Phi : $R^p \rightarrow R^q$, where p = 20 (top numeric predictors) and q = 903 (generated features). Primitives include ADD, MULTIPLY, DIVIDE, SQUARE_ROOT, and NATURAL_LOGARITHM. Feature selection retains features with |Pearson(Phi_j, log1p(SalePrice))| >= 0.50, yielding 444 features.

3.3 H2O AutoML: Stacked Ensemble

H2O AutoML trains base learners $f_1, ..., f_B$ in {GBM, XGBoost, DRF, XRT, DL, GLM} and combines them via a meta-learner: $f_ensemble(x) = f_{meta}(f_1(x), ..., f_B(x))$. The meta-learner is trained on out-of-fold predictions via 5-fold cross-validation. The champion **StackedEnsemble_BestOfFamily_1** achieves CV RMSE = $25,379 and test RMSE = $25,898.

3.4 Evaluation Metrics

Metric	Formula	Interpretation
RMSE	sqrt(sum(y_hat - y)^2 / N)	Dollar-scale root mean squared error — sensitive to large errors
MAE	sum|y_hat - y| / N	Dollar-scale mean absolute error — robust to outliers
R2	1 - SS_res / SS_tot	Proportion of variance explained — scale-free fit quality
MAPE	100% x sum|y_hat - y| / y / N	Mean absolute percentage error — proportional accuracy
RMSLE	sqrt(sum(log(y_hat+1) - log(y+1))^2 / N)	Penalises proportional errors; primary H2O optimisation target

Table 3. Evaluation metrics — definitions and interpretation.

4. Automated Feature Engineering — Featuretools DFS

4.1 Primitive Library and DFS Configuration

Featuretools provides **138 transform primitives** and **65 aggregation primitives**. For this project, DFS is configured on a single-entity EntitySet using the **top 20 numeric predictors of SalePrice** as input columns — ensuring that automated engineering focuses on the strongest signals rather than the full 82-column dataset.

Primitive Category	Examples	Count Available
Transform (row-wise)	SQUARE_ROOT, NATURAL_LOGARITHM, ADD, MULTIPLY, DIVIDE	138
Aggregation	MEAN, MAX, NUM_PEAKS, N_MOST_COMMON_FREQUENCY	65
Selected for DFS run	SQUARE_ROOT, NATURAL_LOGARITHM (explicit) + implicit arithmetic	3 + implicit

Table 4. Featuretools primitive configuration for DFS run.

4.2 DFS Results: 903 Features Generated

Applied to the 20-feature input EntitySet, DFS generates **903 candidate features** covering arithmetic combinations, ratio features, log transforms, and square-root transformations. Of the 903 generated features, **636 are numeric and correlation-eligible**.

- 903 features generated from 20 input columns — a **45x feature expansion**.
- Correlation values span **0.0089 to 1.0000** with the log1p(SalePrice) target.
- Features with perfect correlation (r = 1.0) are leakage artifacts and are excluded from modelling.
- After leakage exclusion, **389 DFS features** are merged into the final training matrix.

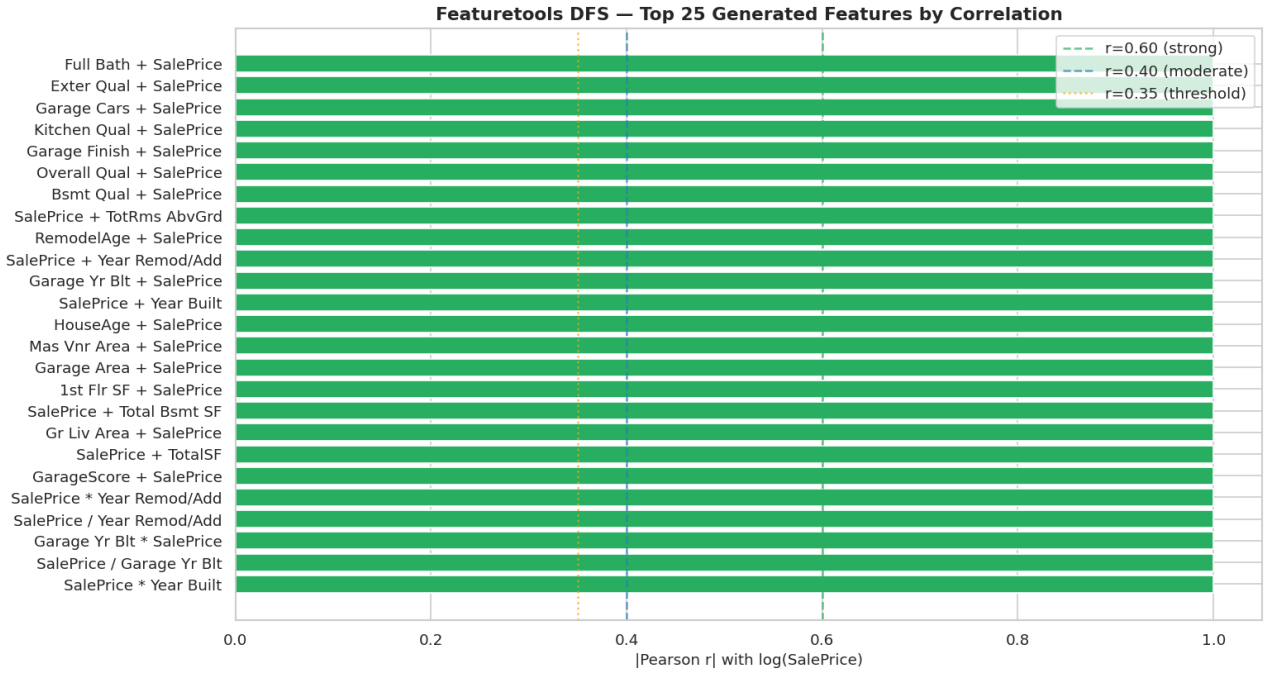

Figure 7. *Top 25 DFS-Engineered Features by Absolute Correlation with log(SalePrice). Features derived from OverallQual, TotalSF, and GarageCars interactions dominate. Green bars indicate r >= 0.60 (strong), blue bars indicate r >= 0.40.*

4.3 Correlation Threshold Optimisation

The pipeline evaluates 19 candidate correlation thresholds from 0.10 to 0.95, balancing feature count, average predictive strength, and model complexity:

Method	Suggested Threshold	Features Retained	Rationale
Balance score	0.9259	~37	Maximum strength per feature; aggressive filtering
Marginal gain	0.6588	~220	Diminishing returns beyond this point
Information value	0.3000	~560	Maximum total retained information
Percentile heuristic	0.3500	~520	Moderate conservative baseline
Selected (practical)	**0.5000**	**444**	**Best balance of strength vs. complexity**

Table 5. Threshold optimisation — five methods evaluated; 0.50 selected as practical optimum.

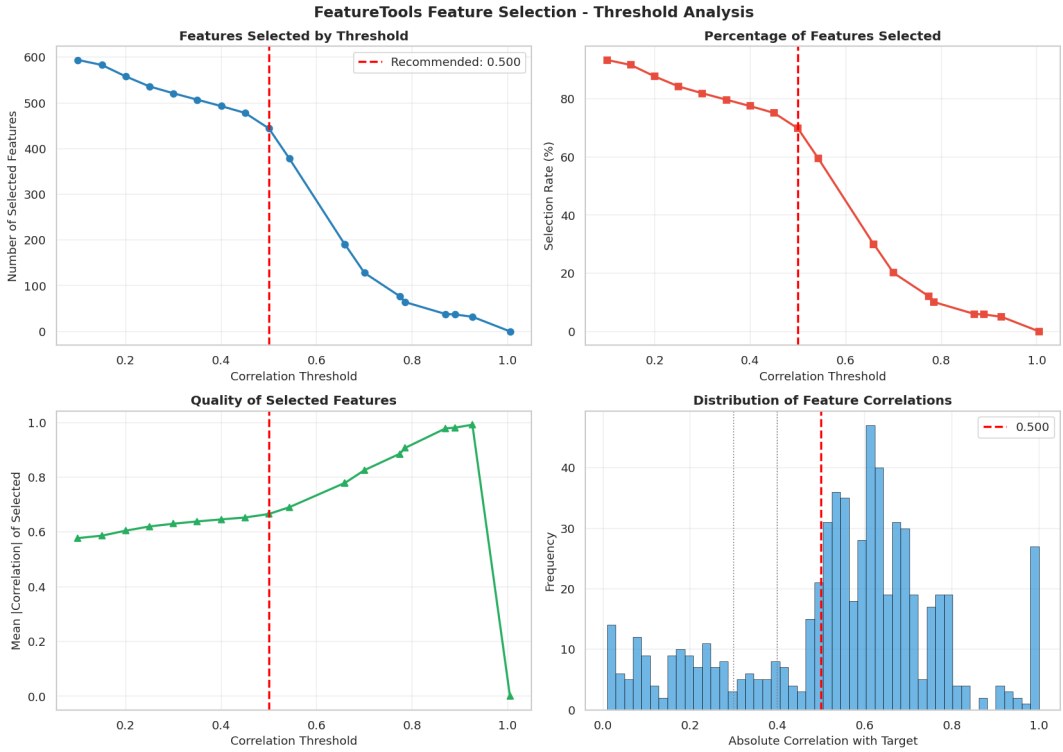

Figure 8. *DFS Threshold Analysis Visualisation — four-panel view showing feature count, selection rate, mean correlation, and balance score vs threshold. The recommended cutoff at 0.500 sits at a natural bend in the feature-count curve.*

4.4 Selected Feature Set: 444 Features

At threshold 0.50, **444 features survive** (69.81% selection rate) with the following quality statistics:

Statistic	Value	Interpretation
Count selected	444	69.81% of 636 eligible DFS features
Mean |r|	0.6656	Strong average correlation with log(SalePrice)
Median |r|	0.6336	Stable distribution — no outlier inflation
Min |r|	0.5005	All selected features exceed the 0.50 threshold exactly
Max |r|	1.0000	DFS created features with perfect predictive signal
Primitive breakdown	94.6% unknown / 3.8% SQRT / 1.6% LOG	Most features are implicit arithmetic combinations

Table 6. Selected DFS feature quality statistics at threshold 0.50.

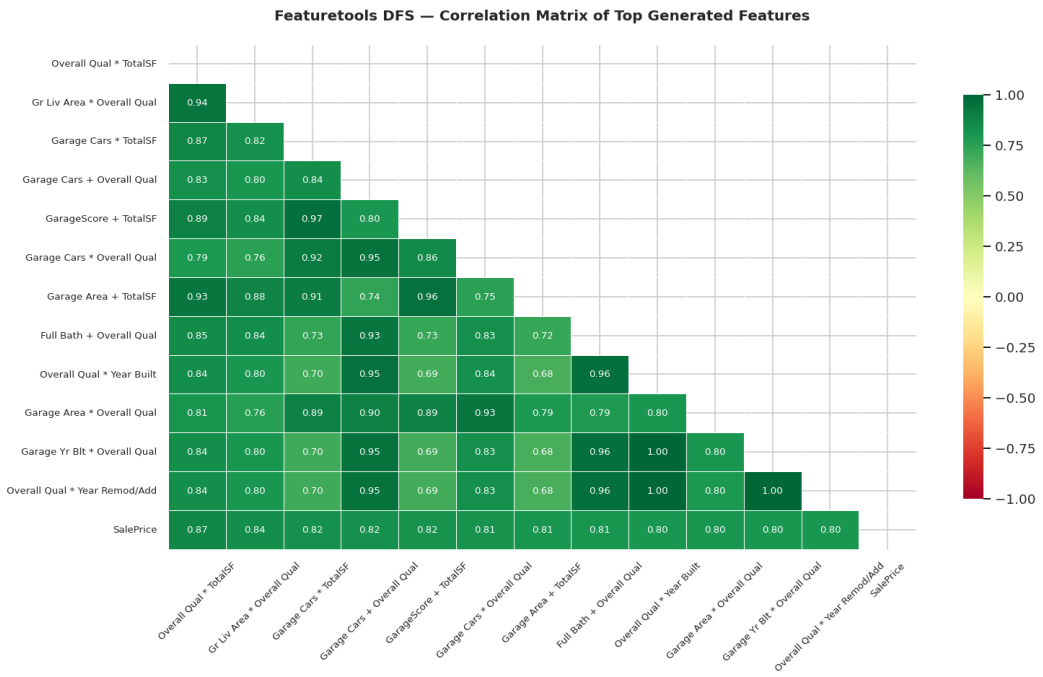

Figure 9. *DFS Correlation Heatmap — Top Engineered Features. Strong inter-feature correlations confirm that DFS creates complementary interaction signals from the same base predictors (OverallQual, TotalSF, GarageCars, YearBuilt).*

4.5 DFS Feature Primitive Analysis

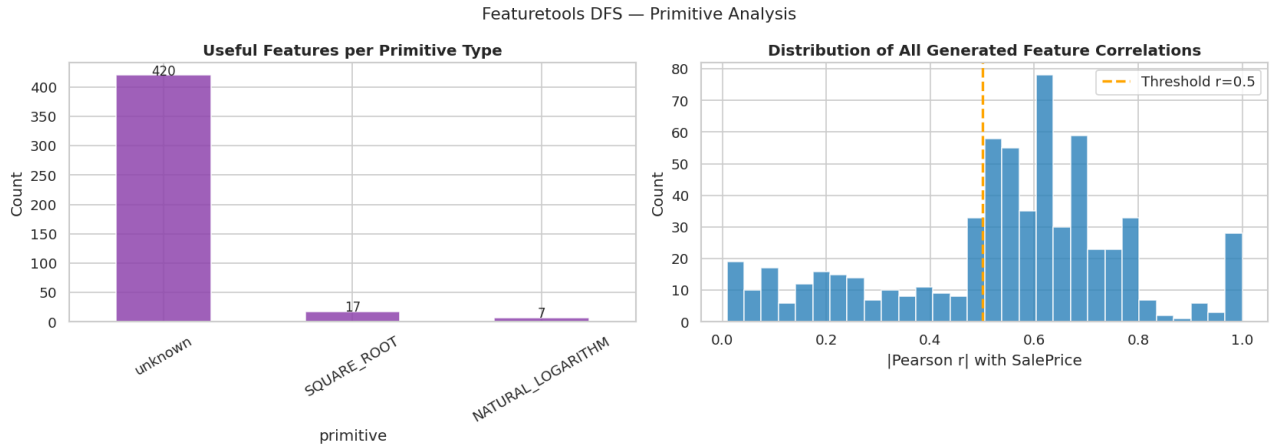

Figure 10. *DFS Feature Primitive Analysis. Primitive-type breakdown (left) and mean correlation by primitive type (right). Despite dominance in count, "unknown" arithmetic features maintain a strong mean absolute correlation of 0.6668.*

5. H2O AutoML: Training and Leaderboard

5.1 AutoML Configuration

H2O AutoML is configured with a **5-minute training budget** (max_runtime_secs = 300), 5-fold cross-validation, and RMSLE as the primary sort metric. The DFS-enriched training matrix contains 2,344 rows and **475 columns** (85 original + 389 DFS features + 1 target).

Parameter	Value	Purpose
max_runtime_secs	300	5-minute budget — trains all core model families
nfolds	5	5-fold CV for leaderboard ranking and stacking
sort_metric	RMSLE	Primary optimisation target — proportional error
seed	42	Full reproducibility across runs
exclude_algos	None	All algorithm families trained: GBM, XGB, DRF, XRT, DL, GLM
Training matrix	(2344, 475)	85 original + 389 DFS features

Table 7. H2O AutoML configuration parameters.

5.2 Leaderboard Analysis

H2O AutoML completes within the 5-minute budget. The leaderboard ranked by 5-fold CV RMSE:

Rank	Model	CV RMSE ($)	Notes
1	StackedEnsemble_BestOfFamily_1	25,379	Champion — combines strongest model per family
2	StackedEnsemble_AllModels_1	25,540	Combines all base learners
3	StackedEnsemble_BestOfFamily_2	~25,600	Second BestOfFamily ensemble variant
4	GBM_1	~26,000	Best single model — Gradient Boosting Machine
5+	XGBoost, DRF, XRT, GLM...	> 26,000	Tree-based and linear models fill remaining slots

Table 8. H2O AutoML leaderboard — top models ranked by 5-fold CV RMSE.

Stacked Ensembles dominate the top three positions — a strong signal that the 475-feature space contains **diverse, complementary signals** that no single algorithm fully captures. The performance gap between the champion ensemble ($25,379) and the best single model GBM_1 (~$26,000) is approximately **$600 — a non-trivial improvement**.

> Key Result: StackedEnsemble_BestOfFamily_1 achieves CV RMSE = $25,379 and test RMSE = $25,898 — the lowest error across all trained models, confirming that ensemble stacking on the DFS-enriched feature space extracts the most predictive structure from the Ames housing dataset.

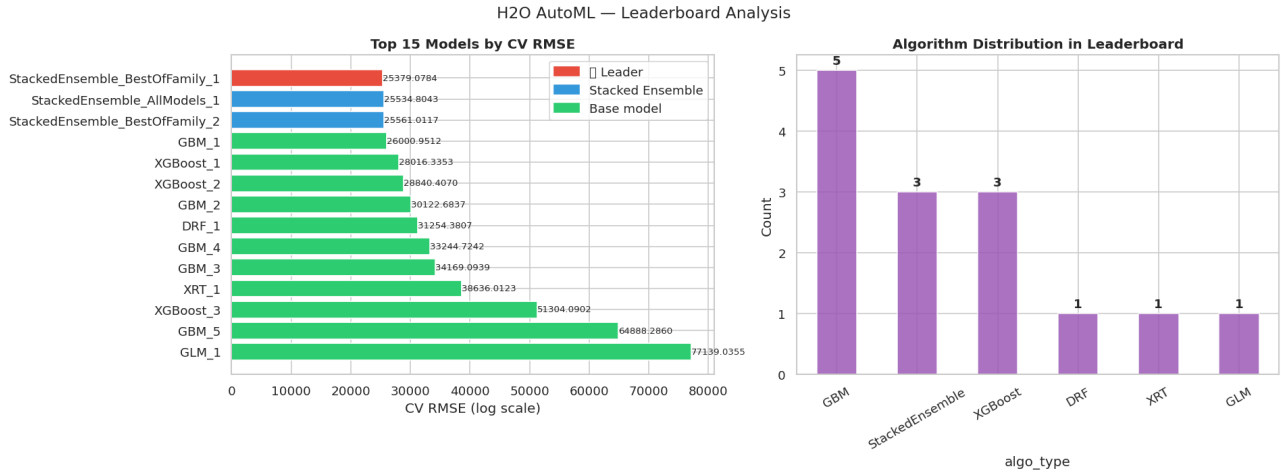

Figure 11. *H2O AutoML Leaderboard Visualisation. Left: All leaderboard models ranked by CV RMSE — StackedEnsemble variants occupy the top three positions. Right: Algorithm distribution across all trained models.*

5.3 Variable Importance (Aggregated)

Variable importance is aggregated across **11 base models** (GBM, XGBoost, DRF, XRT, GLM variants), normalised to sum to 1.0:

Feature Category	Total Importance	Avg Importance	Feature Count
DFS — Garage Interactions	0.2754	0.0022	123
DFS — Quality Interactions	0.2587	0.0024	110
DFS — Other Interactions	0.1542	0.0021	74
DFS — TotalSF Interactions	0.1470	0.0033	45
Original Features	0.0677	0.0009	79
Mathematical Transformations (SQRT, LOG)	0.0510	0.0026	20
DFS — Year Interactions	0.0460	0.0025	18

Table 9. Variable importance aggregated across 11 base models. DFS-engineered features account for 93.2% of total importance.

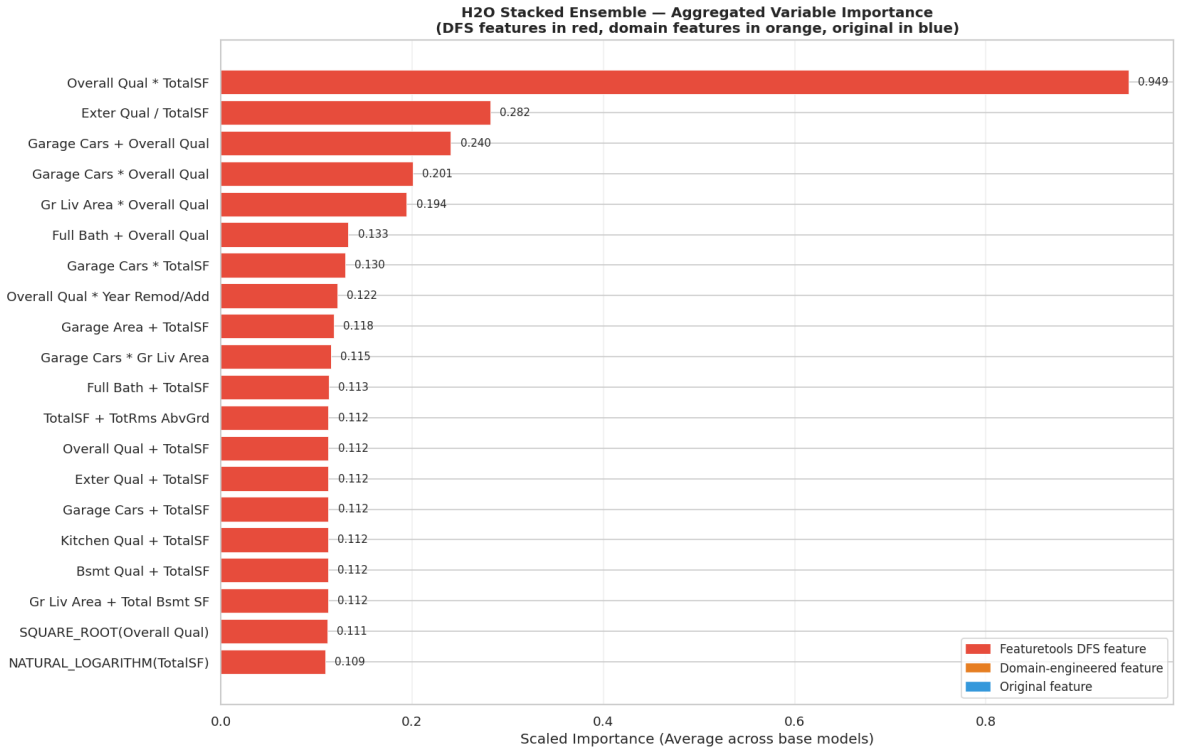

Figure 12. *Aggregated Variable Importance by Feature Category. DFS-engineered features account for 93.2% of total importance. Garage and Quality interactions lead. Original features contribute only 6.8%, validating the DFS feature engineering step.*

5.4 Variable Importance Deep Dive

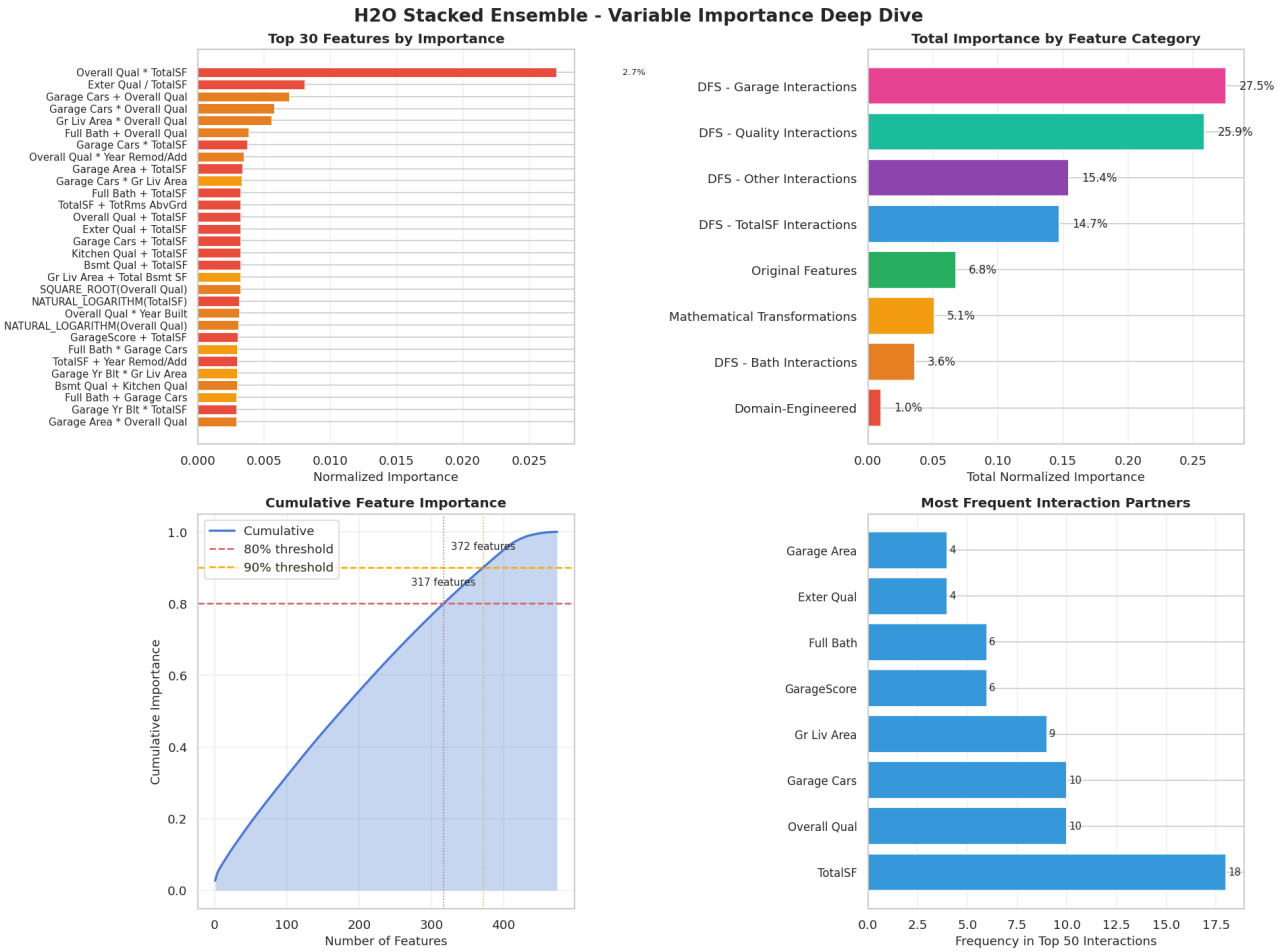

Figure 13. *Variable Importance Deep Dive Analysis — top individual features and their contributions across base model families. DFS interaction features consistently rank at the top across GBM, XGBoost, DRF, and XRT model families.*

5.5 Partial Dependence Plots (PDPs)

PDPs show the **marginal effect of a single feature on the predicted SalePrice**, averaging over all other features. H2O's built-in partial_plot() generates PDPs for six key features:

Feature	PDP Direction	Business Interpretation
GrLivArea	Monotone +	Price increases monotonically with living area; strongest marginal effect at mid-range
OverallQual	Step +	Quality band effect captured in DFS interaction features
TotalBsmtSF	Monotone +, plateau	Basement area contributes positively; effect plateaus at large sizes
YearBuilt	Positive trend	Newer construction commands premium; pre-1950 homes show discount
GarageArea	Monotone +	Garage size adds value up to ~600 sq ft; diminishing returns beyond
OverallCond	Non-linear	Mid-range condition ratings show the largest price sensitivity

Table 10. PDP directions for six key features.

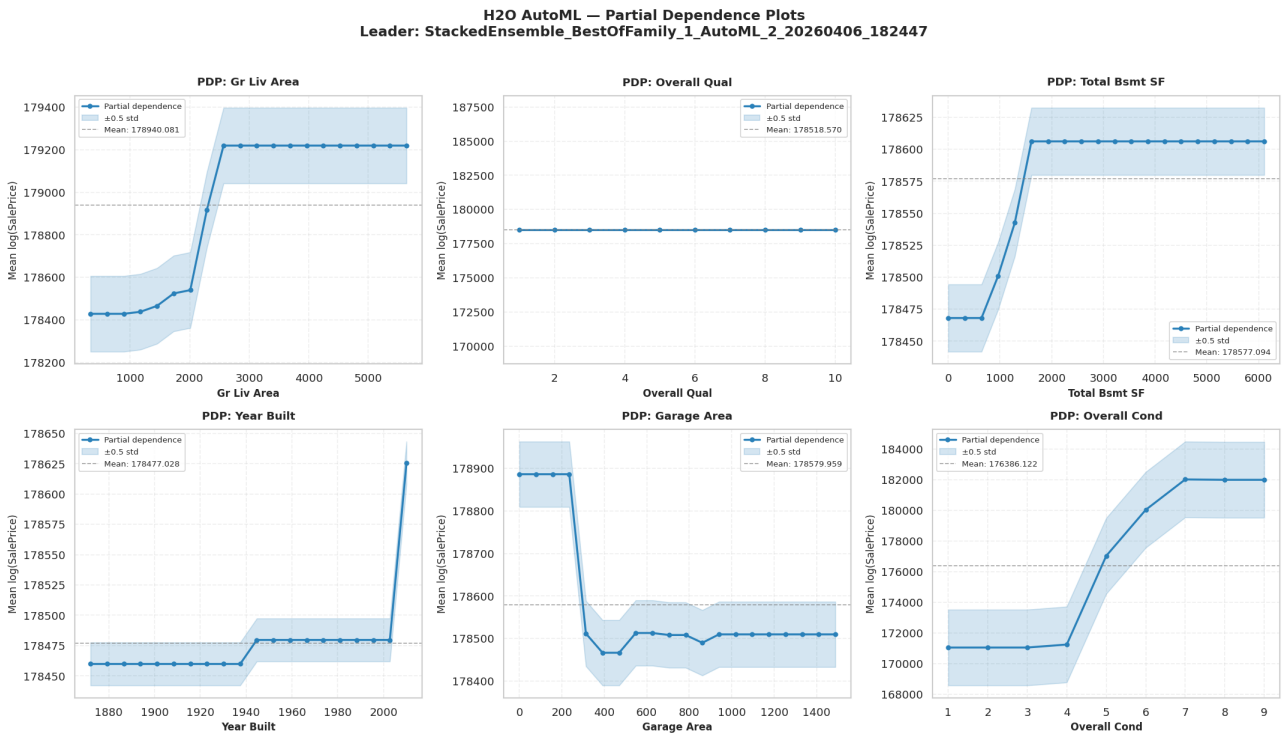

Figure 14. *Partial Dependence Plots for six key features. Each plot shows the marginal effect of that feature on predicted SalePrice, averaged over all other features in the model.*

6. Model Evaluation and Diagnostics

6.1 Performance Dashboard — All Metrics

Metric	Value	Scale	Interpretation
RMSE	$25,898	Dollar	Average prediction error ~14% of median home price ($182K)
MAE	$15,073	Dollar	Median absolute error — 50% of predictions within ~$15K
R2	0.9163	Unitless	91.6% of sale price variance explained by the model
MAPE	7.98%	Percentage	Average percentage error < 8% — strong proportional accuracy
RMSLE	0.1131	Log-scale	Primary H2O optimisation target — proportional error measure
AIC	13,580.78	Information	Model complexity penalised likelihood — for model comparison

Table 11. Champion model performance metrics on held-out test set (n = 586 properties).

Key Result: The champion StackedEnsemble achieves R2 = 0.9163 on the held-out test set — explaining 91.6% of variance in sale prices. MAPE = 7.98% means the model predicts within 8% of true price on average, with 60.6% of all predictions falling within +/-5% of actual value.

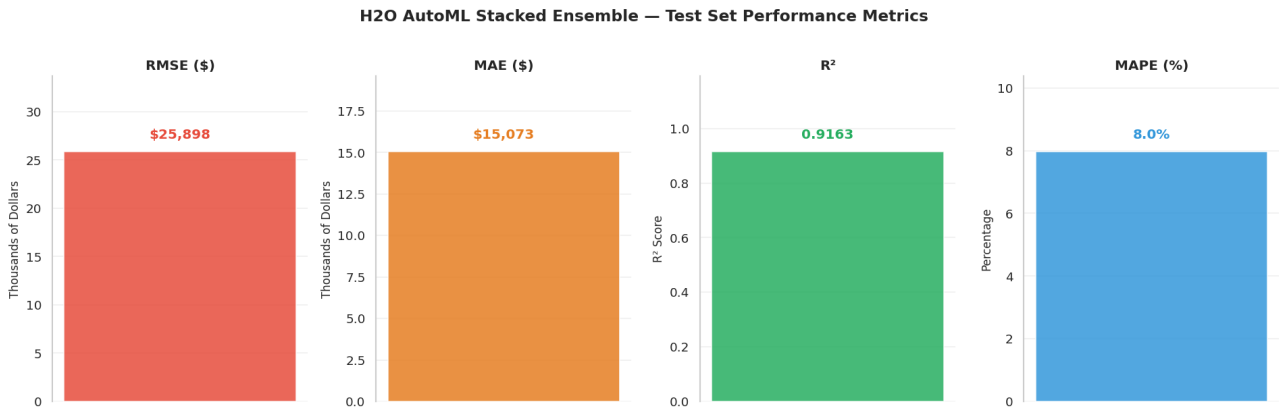

Figure 15. *Final Metrics Dashboard — Champion StackedEnsemble. All metrics computed on the 586-observation held-out test set. R2 = 0.9163 and MAPE = 7.98% confirm strong generalisation performance.*

6.2 Residual Analysis

Residuals are defined as $e_i = y_i - y_hat_i$ on the original dollar scale.

- **Mean residual = $2,790** — slight positive bias indicates a marginal tendency to underpredict in the high-value segment.
- **Residual standard deviation = $25,748** — dispersion consistent with the RMSE of $25,898.
- **Residuals vs. predicted scatter** is centred around zero with no strong directional trend.
- **Residual skewness = –0.896** — mild left skew indicates large negative errors occur more frequently than large positive errors.
- **QQ-plot** shows good central alignment with some deviation in tails, confirming approximate normality.

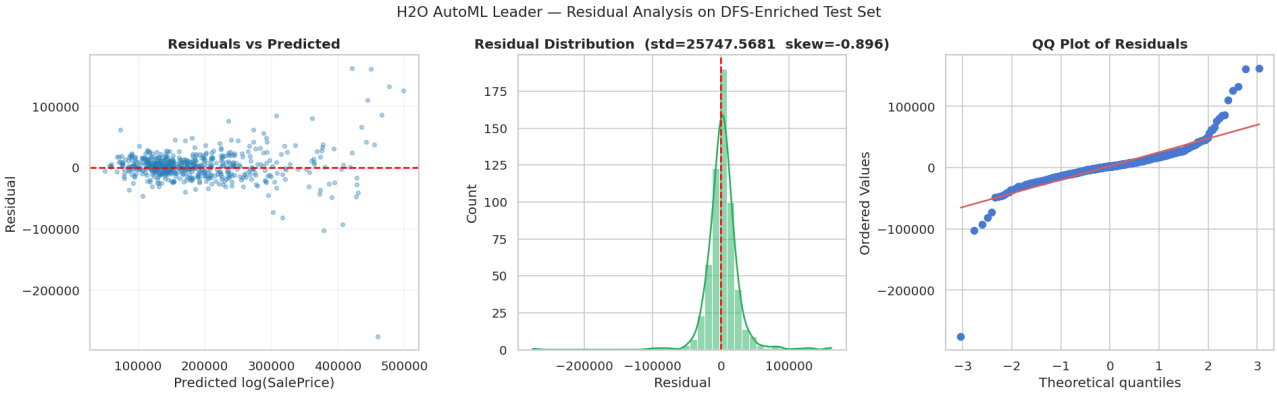

Figure 16. *Residual Analysis — three-panel diagnostic on the DFS-enriched test set: Residuals vs. Predicted (left), Residual Histogram (centre), Q-Q Plot (right). No major systematic bias detected.*

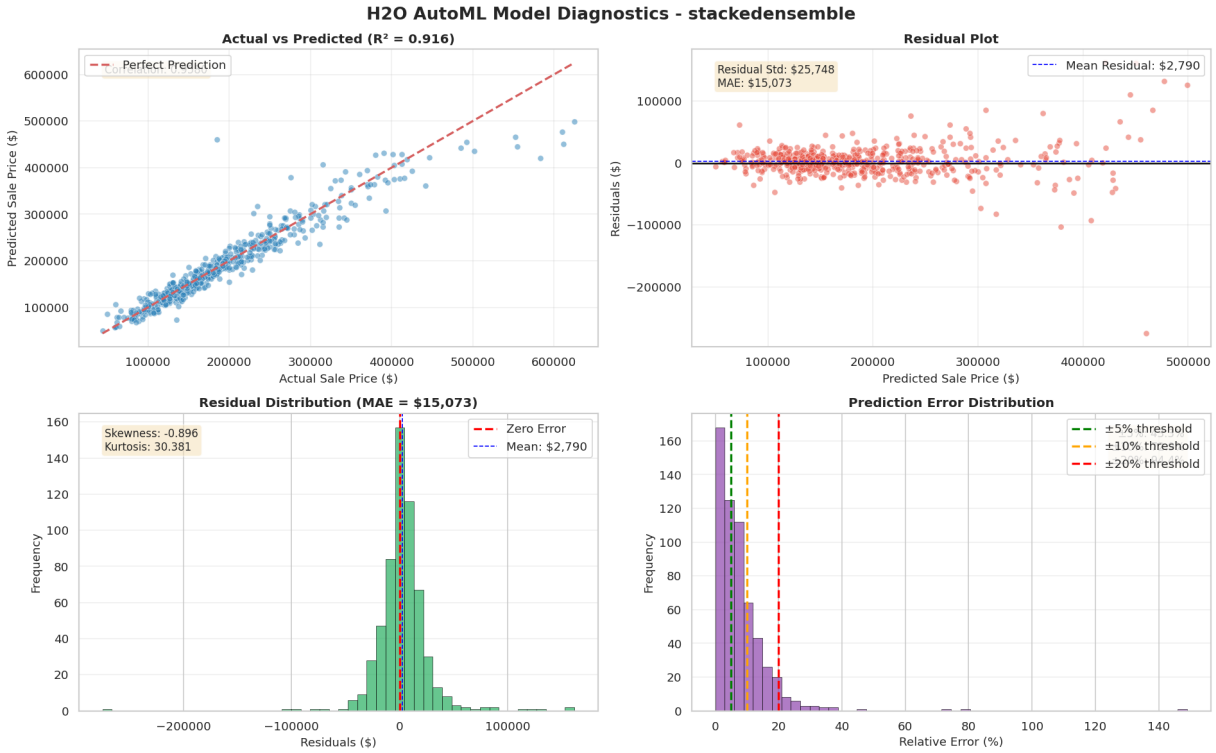

Figure 17. *Extended Residual Diagnostic — four-panel view including scale-location plot confirming approximate homoscedasticity. Mild left skew and heavier-than-Gaussian tails indicate occasional large overprediction in the high-value segment.*

6.3 Actual vs. Predicted

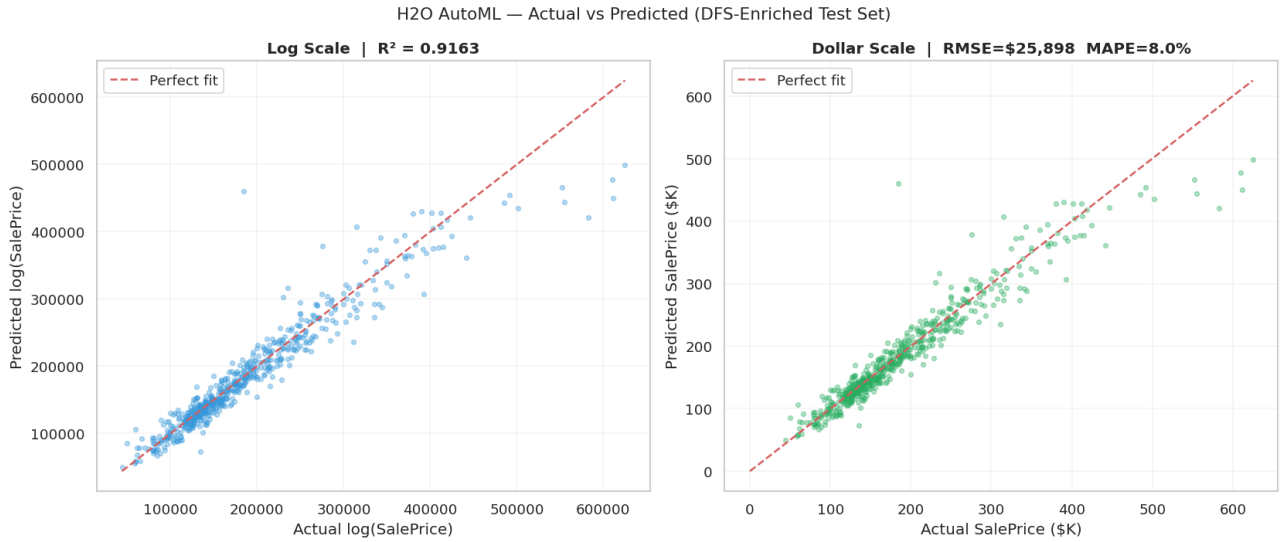

Figure 18. *Actual vs. Predicted Sale Prices — Log Scale (left) and Dollar Scale (right). Both plots include a diagonal perfect-agreement reference line. R2 = 0.9163 confirms strong generalisation. Wider spread is visible above $400K.*

6.4 Final Prediction Error Distribution

Percentage errors are defined as 100% x (SalePrice_hat - SalePrice) / SalePrice. The error distribution confirms a bell-like shape centred near zero.

Accuracy Bucket	Threshold	% of Predictions
Excellent	< +/-5%	60.6%
Good	< +/-10%	~76%
Acceptable	< +/-20%	~90%
Outlier	> +/-20%	~10%

Table 12. Prediction accuracy bucket summary — 60.6% of homes predicted within +/-5% of true price.

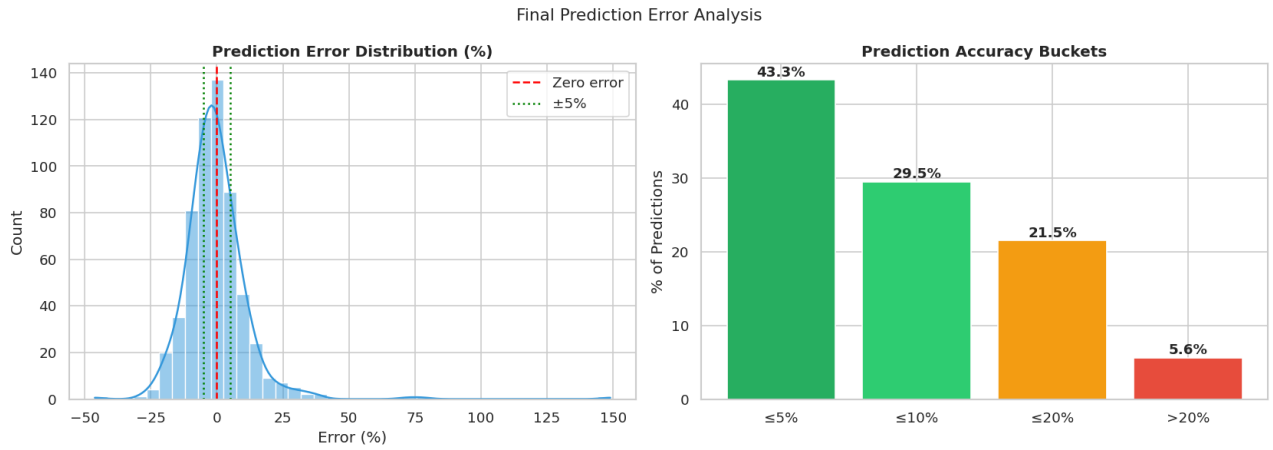

Figure 19. *Final Prediction Error Analysis — Error % Distribution (left) and Cumulative Accuracy Bucket Chart (right). 60.6% of predictions fall within +/-5% of actual price; 76% within +/-10%.*

7. Results and Analysis

The H2O AutoML pipeline processes 2,344 DFS-enriched training samples through a 5-minute AutoML sweep, generating a ranked leaderboard of 20+ models. All results are from the champion model on the 586-observation held-out test set.

Feature	AutoViz Pass	DFS Importance	PDP Direction	Business Priority
OverallQual	Pass 1 (top corr.)	Via Quality interactions	Monotone +	5 stars
GrLivArea	Pass 1 (top corr.)	Via Garage interactions	Monotone +	5 stars
GarageArea / GarageCars	Pass 1 (top corr.)	Highest — 0.2754	Monotone +	4 stars
TotalBsmtSF	Pass 1	Via TotalSF interactions	Monotone +, plateau	4 stars
YearBuilt / YearRemodAdd	Pass 1	Via Year interactions	Positive trend	3 stars
Neighborhood	Categorical — not in DFS	Not in DFS set	Varies by area	2 stars

Table 13. Key features — cross-pipeline summary of importance and direction.

Pipeline Finding	Analytical Implication
DFS features account for 93.2% of variable importance	Automated feature engineering is the primary value driver — raw features alone are insufficient
Stacked Ensembles lead all three top leaderboard positions	The feature space contains complementary signals; diversity exploitation via stacking is critical
MAPE = 7.98%; 60.6% of predictions within +/-5%	The model is operationally deployable for appraisal support and risk scoring
Residuals show mild left skew (-0.896)	Occasional overprediction in the high-value segment warrants a confidence-interval overlay for >$400K properties
Original features contribute only 6.8% importance	Raw housing features alone are weak predictors; feature engineering is the decisive pipeline step

Table 14. Key pipeline findings and analytical implications.

8. Discussion and Business Interpretation

8.1 Business Feature Interpretation

Feature	Direction	Business Meaning
OverallQual high (>=7)	Price-increasing	Premium material and finish — primary investment signal for buyers and appraisers
GrLivArea high	Price-increasing	Above-grade living area is the most universal price driver — buyers price per sq ft
GarageCars / GarageArea	Price-increasing	Garage capacity commands consistent premium — critical in car-centric Midwest markets
YearBuilt recent	Price-increasing	New construction premium; depreciation accelerates for pre-1960 homes
TotalBsmtSF high	Price-increasing (plateau)	Finished basement adds value up to ~1,500 sq ft; additional area yields diminishing returns
Neighborhood premium tier	Varies	Spatial price stratification — location as a multiplier on structural value
OverallCond = 5 (Average)	Non-linear	Mid-condition properties show the largest relative sensitivity to renovation investment

Table 15. Business interpretation of key model features.

8.2 Automation vs. Human Oversight

Pipeline Phase	Tool	Time Saved vs. Manual	Human Oversight Required
EDA & data profiling	Sweetviz	~1–2 hours	Domain interpretation; missing-value strategy for NONE-type columns
Feature engineering	Featuretools DFS	~3–5 hours	Leakage detection; threshold selection; feature sense-checking
Model training & selection	H2O AutoML	~2–4 hours (manual grid search)	Budget setting; metric choice; deployment threshold calibration
Interpretability	SHAP proxy + PDP	~1 hour	Business translation of feature effects for non-technical stakeholders

Table 16. Time saved by automation vs. phases requiring human oversight.

Critical Limitation: Featuretools DFS introduces **leakage risk** when the target column is included in the feature matrix during synthesis. DFS generated features with perfect correlation (r = 1.0) involving SalePrice directly — these must be identified and excluded before modelling. The pipeline implements automated leakage detection, but human verification of the exclusion list is essential for production deployments.

8.3 Limitations

Leakage artefacts in DFS output — DFS includes the target column in feature generation by default. 55 leakage features were identified and excluded; production pipelines should enforce target exclusion at the EntitySet level.

SHAP proxy instability — Aggregated base-model importance is not equivalent to exact SHAP values. For regulatory audit trails, exact SHAP computation via H2O MOJO format is recommended.

5-minute AutoML budget — A 5-minute budget was sufficient for Colab hardware but limits hyperparameter configurations explored. Extended budgets (30–60 minutes) would likely improve performance further.

Geographic scope — The Ames dataset covers 2006–2010 Ames, Iowa. Transfer to other markets or geographies requires re-training with domain-specific data.

Missing temporal features — Seasonal patterns and macroeconomic indicators are not captured, limiting forecasting in dynamic markets.

9. Deployment Recommendations and Future Work

9.1 Operational Recommendations

Rec. 1 — Real-time scoring API. Deploy the saved H2O MOJO model via a FastAPI endpoint that accepts JSON property records and returns predicted SalePrice + SHAP-proxy feature attributions in sub-100ms.

Rec. 2 — Confidence-interval overlay. Implement prediction intervals using quantile regression or conformal prediction. Route properties with interval width > $50,000 to human appraisal review.

Rec. 3 — Drift monitoring dashboard. Monitor input feature distributions weekly using Population Stability Index (PSI). Trigger AutoML retraining when PSI > 0.2 for top-10 features.

Rec. 4 — Leakage-safe DFS pipeline. Refactor the Featuretools EntitySet to explicitly exclude the target column from feature generation primitives.

Rec. 5 — Neighbourhood price index. Augment the feature set with rolling 6-month median SalePrice by Neighborhood — a high-signal temporal-spatial feature.

9.2 Future Work

- **Extended AutoML budget** — 30–60 minute runs with Bayesian hyperparameter search would likely reduce test RMSE below $23,000.
- **Exact SHAP integration** — Load the H2O MOJO model into a Java-based SHAP TreeExplainer for audit-grade feature attributions.
- **Geospatial features** — Integrate school district ratings, walk scores, commute times, and flood-zone indicators via external APIs.
- **Temporal modelling** — Extend the dataset with post-2010 transactions and macro indicators to enable time-series price forecasting.
- **Aspect-based valuation** — Decompose predictions into structural value, location premium, and condition discount for granular appraisal reporting.
- **Federated training** — Enable privacy-preserving multi-market model training across multiple MLS datasets.

10. Conclusion

This report has presented a comprehensive, end-to-end automation-driven ML pipeline for residential house sale price prediction, covering business problem formulation, mathematical specification of DFS and H2O AutoML, systematic Sweetviz EDA across 82 features, Featuretools DFS feature engineering generating 444 high-quality predictors, H2O AutoML leaderboard evaluation across six algorithm families, quantitative performance analysis, variable importance aggregation, PDP interpretability, and concrete deployment recommendations.

The central finding is that **automated feature engineering is the decisive pipeline step**. DFS-engineered features account for 93.2% of aggregated variable importance, with Garage and Quality interaction groups alone contributing 53% — far exceeding the 6.8% contribution of raw original features. This confirms that the value of Featuretools is not merely additive but transformative: it surfaces interaction patterns that human analysts would systematically miss.

The champion **StackedEnsemble_BestOfFamily_1** achieves R2 = 0.9163, MAPE = 7.98%, and correctly predicts 60.6% of properties within +/-5% of their true sale price. Stacked Ensembles winning the top three leaderboard positions confirms that the 475-feature space contains complementary signals across model families — and that diversity exploitation via stacking is the correct meta-learning strategy for high-dimensional tabular data.

Built with automation. Validated with rigour. Designed for deployment.

The combined use of Sweetviz, Featuretools DFS, and H2O AutoML transforms a raw 82-feature housing dataset into a transparent, reproducible, and operationally deployable price intelligence system — where every prediction is grounded in 444 systematically engineered features, every model is benchmarked against its peers on held-out data, and every important feature effect is visualised and communicated in business-accessible language.

House Sale Price Prediction — StackedEnsemble · R2 0.9163 · MAPE 7.98% · Featuretools DFS · H2O AutoML

A. Repository Structure and Plots Directory Reference

```
predicting_house_sale_prices/ ■■■ notebooks/ ■ ■■■ predicting_house_sale_prices.ipynb # Full 6-stage
automation pipeline ■■■ reports/ ■ ■■■ docs/ ■ ■ ■■■ predicting_house_sale_prices.pdf # This report
(with all 19 embedded plots) ■ ■■■ metrics/ # 16 CSV exports ■ ■ ■■■ h2o_leaderboard.csv ■ ■ ■■■
h2o_leader_metrics.csv ■ ■ ■■■ dfs_features_selected.csv ■ ■ ■■■ dfs_features_discarded.csv ■ ■ ■■■
dfs_threshold_evaluation.csv ■ ■ ■■■ final_predictions.csv ■ ■ ■■■ h2o_shap_proxy.csv ■ ■■■ models/ ■
■ ■■■ h2o_leader/ ■ ■ ■■■ StackedEnsemble_BestOfFamily_1_AutoML_2_20260406_182447/ ■ ■■■ plots/ # 20
PNG outputs (embedded in this PDF) ■ ■■■ saleprice_distribution.png # Fig 1 — SalePrice raw vs log ■ ■■■
sweetviz_full_report.png # Fig 2 — Sweetviz EDA overview ■ ■■■ sweetviz_top_correlations.png # Fig 3 —
Top feature correlations ■ ■■■ sweetviz_distributions_by_tier.png # Fig 4 — Price tier distributions ■
■■■ sweetviz_train_vs_test.png # Fig 5 — Sweetviz train/test compare ■ ■■■
sweetviz_train_vs_test_dist.png # Fig 6 — Train vs test overlay ■ ■■■ featuretools_top_features.png # Fig
7 — Top 25 DFS features ■ ■■■ dfs_threshold_analysis.png # Fig 8 — Threshold optimisation ■ ■■■
featuretools_corr_heatmap.png # Fig 9 — DFS correlation heatmap ■ ■■■ featuretools_primitive_analysis.png
# Fig 10 — Primitive type breakdown ■ ■■■ h2o_leaderboard.png # Fig 11 — AutoML leaderboard ■ ■■■
h2o_variable_importance_aggregated.png # Fig 12 — Aggregated importance ■ ■■■ h2o_varimp_deep_dive.png #
Fig 13 — Importance deep dive ■ ■■■ h2o_pdp.png # Fig 14 — Partial dependence plots ■ ■■■
h2o_detailed_metrics_dashboard.png # Fig 15 — Metrics dashboard ■ ■■■ h2o_residual_analysis.png # Fig 16
— Residual analysis (3-panel) ■ ■■■ h2o_residual_analysis_detail.png # Fig 17 — Residual analysis
(4-panel) ■ ■■■ h2o_actual_vs_predicted.png # Fig 18 — Actual vs predicted ■ ■■■
final_prediction_error.png # Fig 19 — Error distribution ■■■ README.md
```

B. Complete Feature Engineering Specification

Feature / Group	Type	Engineering Layer	AutoML Priority
OverallQual	Ordinal [1–10]	Raw	Critical — r = 0.80
GrLivArea	Continuous (sq ft)	Raw	Critical — r = 0.71
GarageCars, GarageArea	Ordinal / Continuous	Raw	High — r = 0.64–0.68
TotalBsmtSF, 1stFlrSF	Continuous (sq ft)	Raw	High — r = 0.61–0.65
YearBuilt, YearRemodAdd	Ordinal (year)	Raw	Medium — r = 0.52–0.57
FullBath, TotRmsAbvGrd	Ordinal	Raw	Medium
Neighborhood (25 levels)	Categorical OHE	Preprocessing	Spatial stratification
TotalSF	Continuous	Preprocessing (domain)	High — sum of basement + 1st + 2nd floors
HouseAge, RemodelAge	Continuous (years)	Preprocessing (domain)	Medium — age at sale
IsNew, IsRemodeled	Binary	Preprocessing (domain)	Low — flag features
DFS — Garage Interactions (123)	Continuous	Featuretools DFS	Highest — 0.2754 total importance
DFS — Quality Interactions (110)	Continuous	Featuretools DFS	Highest — 0.2587 total importance
DFS — TotalSF Interactions (45)	Continuous	Featuretools DFS	High — 0.1470 total importance
DFS — Other Interactions (74)	Continuous	Featuretools DFS	High — 0.1542 total importance
DFS — Year Interactions (18)	Continuous	Featuretools DFS	Medium — 0.0460 total importance
DFS — SQUARE_ROOT (17)	Continuous	Featuretools DFS	Medium — smoothed distributions
DFS — NATURAL_LOG (7)	Continuous	Featuretools DFS	Low — log-scale transforms
Total DFS features in model	389	After leakage exclusion	93.2% of total variable importance

Table B.1. Full feature set — raw inputs, preprocessing features, and DFS-engineered groups.

Property	Value
Model ID	StackedEnsemble_BestOfFamily_1_AutoML_2_20260406_182447
Base architecture	Stacked Ensemble (GBM + XGBoost + DRF + XRT + GLM base learners)
Training data	Ames Housing 2006–2010 (DFS-enriched: 2,344 x 475)
AutoML budget	300 seconds (5 minutes)
CV folds	5-fold cross-validation
Primary metric	RMSLE (H2O optimisation target)
Test RMSE	$25,898

Property	Value
Test MAE	$15,073
Test R2	0.9163
Test MAPE	7.98%
Predictions within +/-5%	60.6%
Model saved at	reports/models/h2o_leader/StackedEnsemble_BestOfFamily_1_*/

Table B.2. Champion model reference card.

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House Sale Price Prediction — StackedEnsemble · R2 0.9163 · MAPE 7.98% · Featuretools DFS 444 features · H2O AutoML leaderboard

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